CSE 715: Computer Graphics

Chapter 2 — Image Representation

Complete Model Answers (2020–2024)

University of Chittagong — 7^{th} Semester B.Sc. (Engg.) Examination

Source: Schaum's Outline of Computer Graphics (Xiang & Plastock), Chapter 2, pp.7–24

Contents

Theoretical Framework — Reusable Across All Years

This section covers the **core theory** that appears in every year's Q1 (and often spills into Q2/Q3). Master these once — every year's Chapter 2 question is a re-mix of the same handful of ideas.

Computer Graphics vs. Image Processing

	Computer Graphics	**Image Processing**
Starting point	An abstract object definition (in object space)	An existing image (in image space)
Operation	Synthesis — object → image	Pixel-based transformation — image → image
Goal	Create a picture from a model	Enhance/analyse/modify a picture that already exists
Example	Rendering a 3D scene from vertex data	Sharpening, denoising, or recoloring a photograph

One-liner (memorise): Computer Graphics goes *object → image*; Image Processing goes *image → image*. Computer–Human Interaction (HCI) is the third allied field — it studies effective communication between user and machine (input devices, GUIs), overlapping with CG mainly at graphical user interfaces.

How a digital image is formed from an object definition (the graphics pipeline)

Object Definition (object space) → Transformation (world coords) → Scan Conversion (image/pixel space)

A shape (e.g. a triangle's vertex coordinates) is defined abstractly in **object space** (continuous), placed/scaled/rotated into the **world coordinate system** via **transformation** (matrices), then **scan-converted** (rasterized) into discrete pixels in **image space**.

The RGB and CMY Color Models

- **RGB (additive):** Start from **black** $(0, 0, 0)$, *add* Red/Green/Blue light to reach the desired color. White $= (1, 1, 1)$. Matches how a **display monitor** works (emits light).
- **CMY (subtractive):** Start from **white** $(0, 0, 0)$ in CMY-space, *subtract* (absorb) primaries using Cyan/Magenta/Yellow pigment to reach the desired color. Black $= (1, 1, 1)$ in CMY. Matches how a **printer** works (pigments absorb light reflected off paper).

- **Conversion formulas:**

$$\begin{pmatrix} R \\ G \\ B \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} C \\ M \\ Y \end{pmatrix} \qquad \begin{pmatrix} C \\ M \\ Y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} R \\ G \\ B \end{pmatrix}$$

- **Why printers add a 4th, black (K) pigment (CMYK):** producing a deep, pure black by mixing C+M+Y is technically difficult (tends toward muddy brown) and **more expensive** (uses 3 costly color pigments instead of 1 cheap black one) than using a dedicated black ink.

Perceptual color terms vs. physical light properties

Perceptual term	Physical property of light
Hue	Dominant wavelength
Saturation (Purity)	Excitation purity — how "undiluted" by white light the color is
Brightness (Lightness/Value)	Luminance / intensity

Direct Coding

Direct coding allocates a fixed number of bits per pixel to code its color **directly** — no indirection. If n_r, n_g, n_b bits are used for R, G, B respectively:

$$\text{number of colors per pixel} = 2^{n_r} \times 2^{n_g} \times 2^{n_b}$$

If all three primaries use the same n bits: colors $= (2^n)^3 = 2^{3n}$.

Special cases: **bilevel** (black/white) images need only 1 bit/pixel; **gray-scale** images typically use 8 bits/pixel (256 levels), coded once since R=G=B. The industry-standard **24-bit "true color"** format uses 1 byte (8 bits, 256 levels) per primary $\Rightarrow 256^3 = 16.7$ million colors.

Lookup Table (LUT)

Instead of coding color directly, each pixel stores an **address/index** into a table of colors (the *color map*). An n-bit pixel value indexes a table of 2^n entries; each entry typically holds a full 24-bit RGB color.

$$\text{LUT entries} = 2^{(\text{bits per pixel})} \qquad \text{LUT storage (bits)} = 2^{(\text{bits per pixel})} \times (\text{bits per entry})$$

Trade-off: LUT gives a smaller storage footprint (e.g. 8-bit pixel + 24-bit table = "8-bit format", 256 simultaneous colors chosen from 16.7M) at the cost of only 2^n *simultaneous* colors in one image, vs. direct coding's full color range being available pixel-by-pixel.

Aspect Ratio, Resizing, and Sub-Image Cropping

Aspect ratio $= \dfrac{\text{width}}{\text{height}}$ (unit length or pixels).
Resize without distortion requires the *new* aspect ratio to equal the *original* aspect ratio: $\dfrac{w_1}{h_1} = \dfrac{w_2}{h_2}$. If they differ, the image is stretched/squeezed — **geometric distortion** occurs.
Center-crop a $w \times h$ sub-image from a $W \times H$ image: lower-left corner of the crop in the big image $= \left(\dfrac{W-w}{2}, \dfrac{H-h}{2}\right)$; upper-right corner = lower-left $+ (w, h)$.

Halftoning, Halftone Approximation, and Dithering

- **Halftoning:** a printing-industry technique using **variably-sized** pigment dots (arranged at a 45° screen angle) that, viewed from a distance, blend with the white background to simulate continuous-tone shading on a **bilevel** device. Dot size $\propto$ intensity level (bigger dot $\rightarrow$ darker perceived tone).
- **Halftone approximation:** instead of varying dot *size*, use an $n \times n$ grid of **fixed-size** bilevel (or multi-level) pixels turned on in a specific growth pattern to approximate the look of variable dot sizes.

With an $n \times n$ pixel grid where each pixel can take m intensity levels:

$$\text{overall intensity levels} = n \times n \times (m-1) + 1$$

(the "+1" accounts for the all-off / zero-intensity level.)

- **Dither matrix:** an $n \times n$ matrix D_n that encodes the *order* in which grid pixels are turned on. Pixel (x, y) is turned on if the image's intensity level at that position exceeds $D_n(i, j)$ where $i = x \bmod n$, $j = y \bmod n$ (the matrix "tiles" across the whole image like floor tiles — this is called **dithering**, and it approximates halftones *without* sacrificing spatial resolution).
- **Recurrence relation to build a bigger dither matrix from a smaller one** (Bayer matrix construction):

$$D_{2n} = \begin{pmatrix} 4D_n & 4D_n + 2 \\ 4D_n + 3 & 4D_n + 1 \end{pmatrix}$$

Examination 2024 — Q2, Q3 (Section A)

Question breakdown:

Q2(a) Explain the concept of a color model and why RGB/CMY are both used. (3)

Q2(b)	Crop 128×128 from the center of a 512×512 image — upper-left corner coords. (2)
Q2(c)	10-bit grayscale pixel values in a LUT — entries required. (1)
Q2(d)	What is halftoning; how does it create continuous-tone appearance. (3)
Q3(a)	Given a halftone-pattern image, describe how dot size/spacing changes create intensity perception. (2)
Q3(b)	Given $D_2 = \begin{pmatrix} 0 & 2 \\ 3 & 1 \end{pmatrix}$ and the recurrence, construct D_4. (3)

Q2(a) — Color Model Concept [3 marks]

See §1.2 above. **Key point:** RGB matches *emissive* devices (monitors, additive mixing of light); CMY matches *reflective* devices (printers, subtractive absorption of light from ambient/incident light). Both are needed because graphics content must round-trip between screen (RGB) and print (CMY).

Q2(b) — Crop Coordinates [2 marks]

$$\text{Upper-left corner in the big image} = \left(\frac{512-128}{2}, \frac{512-128}{2}\right) = (192, 192).$$

Q2(c) — LUT Entries [1 mark]

10-bit pixel values $\Rightarrow 2^{10} = \mathbf{1024}$ entries required.

Q2(d) — What is Halftoning [3 marks]

See §1.6. Halftoning uses variable dot size at a fixed grid pitch; larger dots reflect less white paper → perceived as darker; smaller dots → perceived as lighter. Viewed from normal reading distance, the eye cannot resolve individual dots and integrates them into a smooth gray-level gradient.

Q3(a) — Halftone Pattern Description [2 marks]

As intensity increases across the gradient, dot **size grows** (occupying more of each cell) while center-to-center **spacing (pitch) stays fixed** — larger dots merge visually, reducing the fraction of visible white background, which the eye reads as a smooth transition from light to dark.

Q3(b) — Construct D_4 from D_2 [3 marks]

$$D_2 = \begin{pmatrix} 0 & 2 \\ 3 & 1 \end{pmatrix}, \qquad D_4 = \begin{pmatrix} 4D_2 & 4D_2 + 2 \\ 4D_2 + 3 & 4D_2 + 1 \end{pmatrix}$$

$$4D_2 = \begin{pmatrix} 0 & 8 \\ 12 & 4 \end{pmatrix}, \quad 4D_2{+}2 = \begin{pmatrix} 2 & 10 \\ 14 & 6 \end{pmatrix}, \quad 4D_2{+}3 = \begin{pmatrix} 3 & 11 \\ 15 & 7 \end{pmatrix}, \quad 4D_2{+}1 = \begin{pmatrix} 1 & 9 \\ 13 & 5 \end{pmatrix}$$

$$D_4 = \begin{pmatrix} 0 & 8 & 2 & 10 \\ 12 & 4 & 14 & 6 \\ 3 & 11 & 1 & 9 \\ 15 & 7 & 13 & 5 \end{pmatrix}$$

(This is the well-known 4×4 **Bayer ordered-dither matrix** — recognise it on sight.)

Examination 2023 — Q1 (Section A, parts a/b/g of 9)

Question breakdown (Q1 is 9×1 = 9 marks total, a–i):

(a) How can you differentiate computer graphics from image processing? (1)
(b) Crop a 786×660 sub-image from the center of a 1200×1200 image — lower-left & upper-right pixel coords. (1)
(g) Name the three perceptual terms for describing color & the corresponding physical property of light. (1)

(Parts c–f, h, i of Q1 2023 belong to other topics — projection/hidden-surface/color-transform clusters; not repeated here.)

Q1(a) — CG vs. Image Processing [1 mark]

See §1.1 — CG: object → image; Image Processing: image → image.

Q1(b) — Crop Coordinates [1 mark]

Lower-left $= \left(\dfrac{1200-786}{2}, \dfrac{1200-660}{2}\right) = (207, 270)$.
Upper-right = lower-left $+(786, 660) = (993, 930)$.

Q1(g) — Perceptual Color Terms [1 mark]

Hue ↔ dominant wavelength; **Saturation** ↔ purity of the color (how little it's diluted by white); **Brightness** ↔ luminance/intensity of the light.

Examination 2022 — Q1 (Section A, parts a–f)

Question breakdown:

(a) How do CG, Image Processing, and Human-Computer Interaction differ? (implicit within Q1)
(b) Resize a 1680×1050 image to 1024 wide, same aspect ratio — new height? (2)
(c) 2-byte pixel values in a LUT representation — bits occupied? (1)
(d) Direct coding of CMY: 3 bits cyan, 3 bits magenta, 4 bits yellow — possible colors per pixel? (1)
(e) Can a 5×3.5in image be shown at 3.5×4in without geometric distortion? (2)
(f) Why do color printers use a subtractive color model? (1)

Q1(b) — Resize Preserving Aspect Ratio [2 marks]

height $= 1024 \times \dfrac{1050}{1680} = \mathbf{640}$ pixels.

Q1(c) — LUT Bits Occupied [1 mark]

2-byte (16-bit) pixel $\Rightarrow 2^{16} = 65{,}536$ entries. If each entry is a standard 24-bit RGB value: $65{,}536 \times 24 = \mathbf{1{,}572{,}864}$ bits (= 196,608 bytes).

Q1(d) — Direct Coding CMY, Mixed Bit Widths [1 mark]

$2^3 \times 2^3 \times 2^4 = 8 \times 8 \times 16 = \mathbf{1024}$ possible colors.

Q1(e) — Distortion Check [2 marks]

Original AR $= 5/3.5 = 1.4286$. Target AR $= 3.5/4 = 0.875$. Not equal $\Rightarrow$ **No**, distortion occurs (the image will appear stretched/squeezed).

Q1(f) — Why Subtractive Model for Printers [1 mark]

Printers deposit pigment that **absorbs** certain wavelengths of ambient/reflected light rather than emitting light directly — this passive, reflective process is inherently subtractive (start from white paper, subtract color), unlike a monitor which actively emits RGB light (additive).

Examination 2021 — Q1 (Section A, parts a–e)

Question breakdown:

(a) How does image processing differ from computer graphics? Briefly explain. (1)
(b) 2-byte pixel values in a 24-bit LUT representation — bits occupied? (1.25)
(c) Direct coding RGB, 12 bits per primary color — possible colors per pixel? (1)
(d) Resize a 1024×675 image onto two devices: 800×527 and 1800×1100 — what phenomena occur, with computation. (3)
(e) Crop a 700×700 sub-image from the center of a 900×800 image — upper-right corner coords. (1.5)

(Part f, the Koch curve, belongs to the Scan Conversion/Fractals topic — covered separately.)

Q1(b) — LUT Bits Occupied [1.25 marks]

$2^{16} \times 24 = \mathbf{1{,}572{,}864}$ bits.

Q1(c) — Direct Coding, 12 bits/primary [1 mark]

$2^{12} \times 2^{12} \times 2^{12} = 2^{36} = \mathbf{68{,}719{,}476{,}736}$ possible colors.

Q1(d) — Resize onto Two Devices [3 marks]

Original AR = 1024/675 = 1.51704.
Device (a) 800 × 527: AR = 800/527 = 1.51803 — within $\sim$ 0.07% of original: **negligible distortion**, image displays essentially correctly.
Device (b) 1800×1100: AR = 1800/1100 = 1.63636 — roughly 7.9% off from original: **noticeable geometric distortion** (image appears stretched wider relative to its height).

(a) Near-zero distortion (AR essentially matches). **(b)** Visible distortion — the picture looks stretched because the device's aspect ratio doesn't match the image's.

Q1(e) — Crop Coordinates [1.5 marks]

Lower-left $= \left(\dfrac{900-700}{2}, \dfrac{800-700}{2}\right) = (100, 50)$.
Upper-right = lower-left $+(700, 700) = (800, 750)$.

Examination 2020 — Q1, Q2 (Group-A, parts a–d and a–c)

Question breakdown:

Q1(a) Which discipline describes producing and synthesizing digital images? Justify. (3)
Q1(b) Can a 5×3.5in image be presented at 3.5×4in without geometric distortion? (2)
Q1(c) How can a digital image be formed from an object definition? (1.25)
Q1(d) Why is a standard aspect ratio necessary? Height = 2in, AR= 1.5 — find width. (1+1.5)
Q2(a) What do you mean by subtractive color model? Give an example. Why do printers use a separate black cartridge? (1+1)
Q2(b) Direct coding RGB, 10 bits/primary — possible colors per pixel? (1.5)
Q2(c) Crop 700×700 from the center of a 900×800 image — upper-right corner coords. (2.75)

(Q2(d), flood-fill, belongs to the Scan Conversion topic — covered separately.)

Q1(a) — Discipline Identification [3 marks]

Computer Graphics. Justification: it is the field concerned with generating and synthesizing digital images from geometric/object definitions, as opposed to capturing images from the real world (photography) or processing existing images (image processing).

Q1(b) — Distortion Check [2 marks]

AR original $= 5/3.5 = 1.4286$; AR target $= 3.5/4 = 0.875$. Not equal $\Rightarrow$ distortion occurs.

Q1(c) — Digital Image Formation [1.25 marks]

See the pipeline diagram in §1.1: Object Definition → Transformation → Scan Conversion.

Q1(d) — Aspect Ratio Necessity + Width Calc [1+1.5 marks]

A standard aspect ratio must be maintained so images display without geometric distortion across devices of different physical dimensions/resolutions.

Width = AR $\times$ height = $1.5 \times 2 =$ **3** inches.

Q2(a) — Subtractive Color Model [1+1 marks]

Subtractive model: start from white, *subtract* (absorb) primary components using pigment to reach a color; e.g. **CMY** (cyan, magenta, yellow) used by printers. **Why a separate black cartridge:** mixing full C+M+Y to make black is costly (three expensive pigments) and technically hard to get a true, deep black (tends muddy) — a dedicated black (K) ink is cheaper and gives cleaner text/blacks.

Q2(b) — Direct Coding, 10 bits/primary [1.5 marks]

$2^{10} \times 2^{10} \times 2^{10} = \mathbf{1{,}073{,}741{,}824}$ possible colors.

Q2(c) — Crop Coordinates [2.75 marks]

Lower-left $= (100, 50)$, upper-right $= (800, 750)$ — identical setup to 2021 Q1(e) above.

Quick Summary — Exam Patterns for Chapter 2

Pattern	**Years**	**Formula / Method to memorise**
Center-crop sub-image coords	every year	lower-left $= \left(\frac{W-w}{2}, \frac{H-h}{2}\right)$, upper-right $=$ lower-left $+(w,h)$
Direct coding bit → color count	every year	colors $= 2^{n_r} \times 2^{n_g} \times 2^{n_b}$
LUT entries/bits	every year	entries $= 2^{\text{bits/pixel}}$; storage = entries × bits/entry
Aspect-ratio distortion check	every year	compare w_1/h_1 vs. w_2/h_2 — equal = no distortion
CG vs. Image Processing	every year	object→image (CG) vs. image→image (IP)
Subtractive/CMY reasoning	2020, 2022, 2023	printers absorb light (subtractive); K added for cost + true black
Halftoning & dither matrix	2024 only (new)	recurrence $D_{2n} = \begin{pmatrix} 4D_n & 4D_n+2 \\ 4D_n+3 & 4D_n+1 \end{pmatrix}$

Exam-day strategy for Q1/Q2 (Chapter 2 cluster): this is almost always a fast, low-risk 8–13 marks if you drill the five formulas above cold — every sub-part is a 1–3 mark plug-and-compute, not a derivation. Spend no more than 15–20 minutes of exam time here; bank the marks early and move to the heavier Ch4/Ch5/Ch7/Ch10 questions.